Visualizing and Measuring Reality

Lecture 2: Descriptive Statistics: Graphical and Numerical Techniques (Chapters 2 & 3)

Hamid Bekamiri
March 11

Textbook: Keller, G. & Gaciu, N. (2020). Statistics for Management and Economics, 2nd ed., Cengage, ISBN: 9781473768260.

Recap - The Big Picture

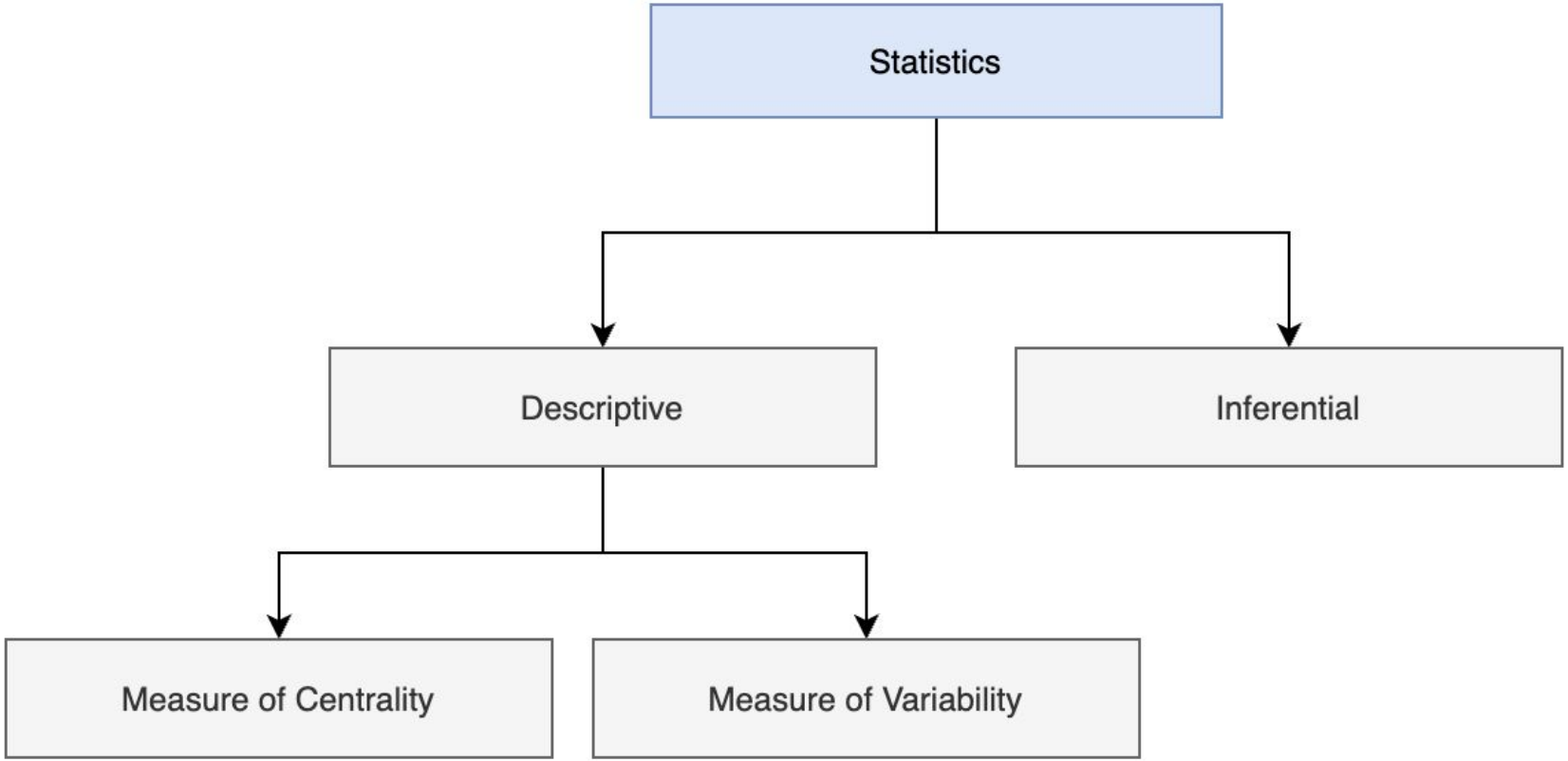
- Remember our map of Statistics from last week?
- We are firmly in the realm of **Descriptive Statistics**: gathering data about a group to summarize and reach conclusions about that *same* group.
  - **Chapter 2:** Summarizing with Shapes and Colors (Graphical).
  - **Chapter 3:** Summarizing with Hard Math (Numerical).

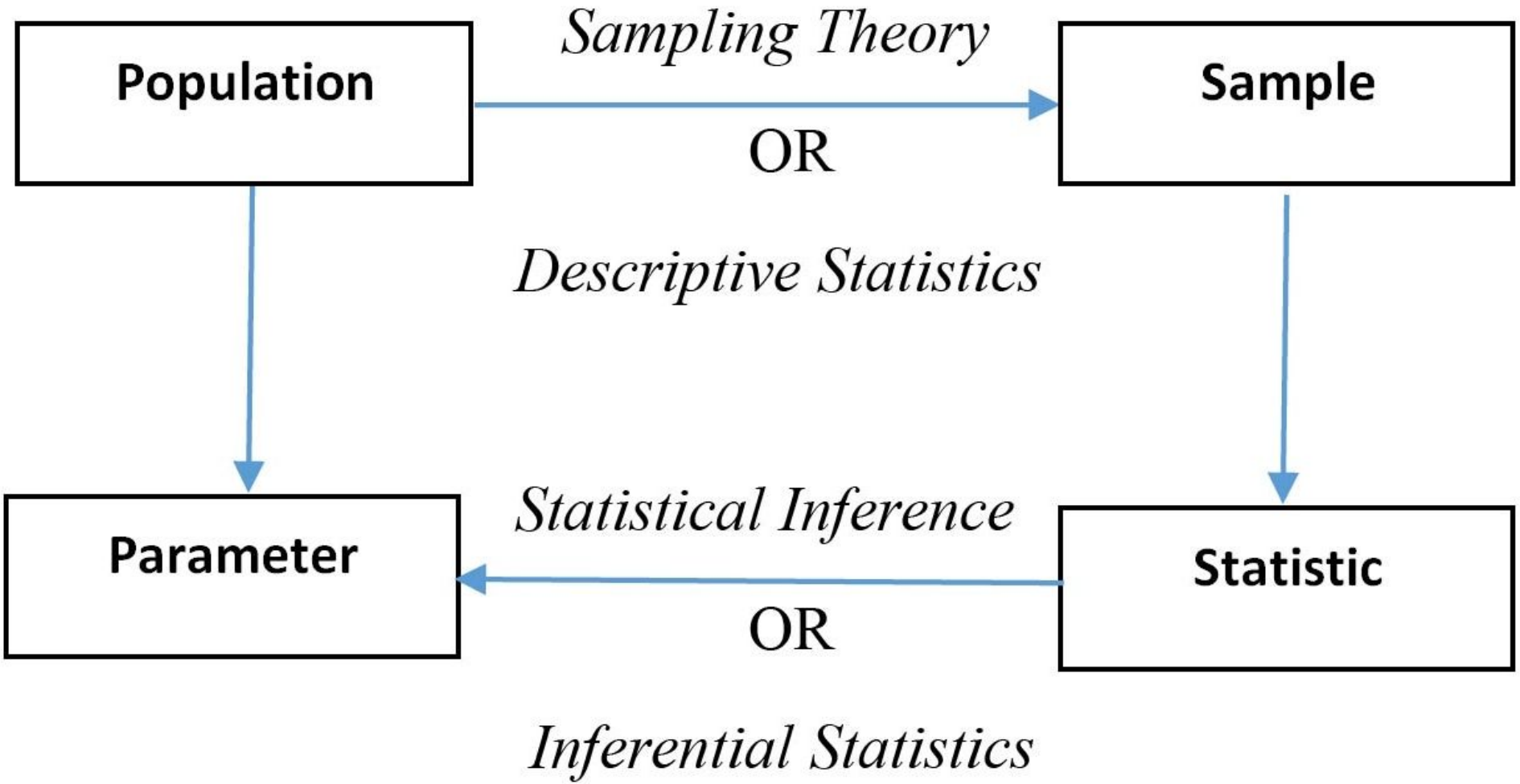
The Goal: Turn raw data into actionable business information.

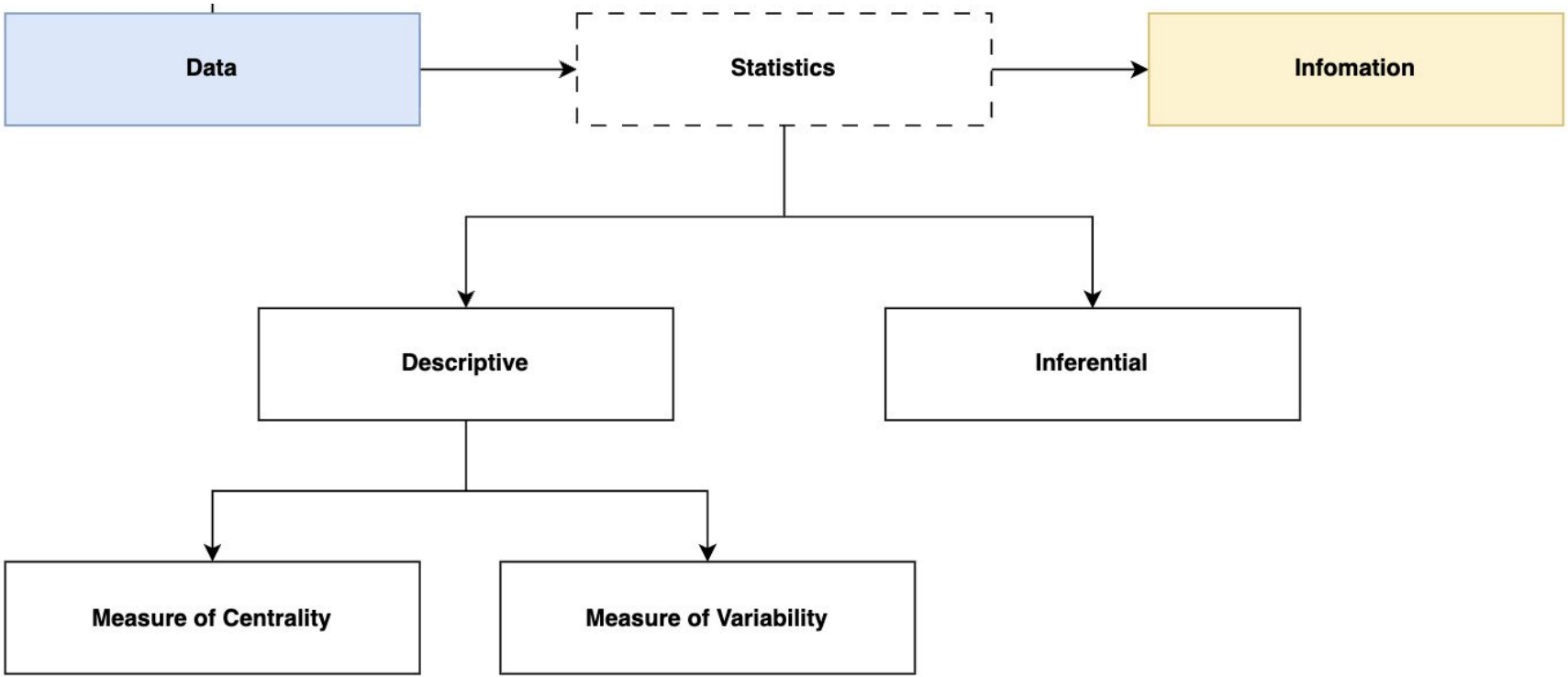
What is Statistics?

Data: Facts, especially numerical facts, collected together for reference or information.
Information: Knowledge communicated concerning some particular fact.

Statistics is a **tool** for creating **new understanding** from a set of numbers.


Statistics
Descriptive
Inferential
Measure of Centrality
Measure of Variability


Population
Sampling Theory
Sample
OR
Descriptive Statistics
Parameter
Statistical Inference
Statistic
OR
Inferential Statistics


Data
Statistics
Infomation
Descriptive
Inferential
Measure of Centrality
Measure of Variability

WHY STATISTICAL VISUALIZATION?

Visualization is a fantastic way to communicate information from data

Top executives make decisions 25% faster when they are presented with materials that include graphics – *Wharton School of Business*

HOW DO WE LEARN VISUALIZATION?

"We learn statistics by doing statistics" – *Keller & Gaciu 2020*

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What is Data?

Data: Facts, especially numerical facts, collected together for reference or information.
Information: Knowledge communicated concerning some particular fact.

Data are ***the observed values*** of a ***variable*** e.g. grades: {7, 7, 4, 0, 10, 7, 12}

A **variable** is a piece of information about a sample or a population e.g. grades

Values of a variable is the range of possible values the variable can take e.g. grades (-3..12)

CASEID	INCOME	AGE	RACE	MARITAL	CHILDS
57062	82.500	53	1	3	0
57063	175.000	26	1	1	0
57064	45.000	59	1	3	1
57065	175.000	56	1	1	2
57066		74	1	1	3
57067	175.000	56	1	1	1
57068		63	1	1	2
57069	82.500	34	1	1	2
57070	16.250	37	1	5	4
57071	100.000	30	3	1	3
57072	11.250	43	3	1	2
57073	11.250	56	1	5	0
57074	18.750	69	1	3	5
57075	23.750	40	1	1	2
57076	27.500	25	1	5	0
57077	67.500	56	1	1	3
57078	82.500	51	3	3	3
57079	175.000	46	1	5	0
57080	27.500	51	1	1	2
57081	7.500	39	3	3	2
57082	23.750	30	2	3	2
57083		36	1	5	3
57084	11.250	42	1	4	3
57085	32.500	38	3	5	0
57086	32.500	38	1	5	2
57087	500	28	2	5	0
57088	23.750	35	1	4	2
57089	120.000	57	1	1	6
57090	175.000	50	1	1	2
57091	45.000	54	1	3	1
57092	100.000		1	1	0

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57089	120.000	57	1	1	6
57090	175.000	50	1	1	2
57091	45.000	54	1	3	1
57092	100.000		1	1	0

For each row there is an observation

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57092	100.000		1	1	0

For each column there is a variable

Know Your Data Types Before You Code

Categorical Data: Qualitative labels.

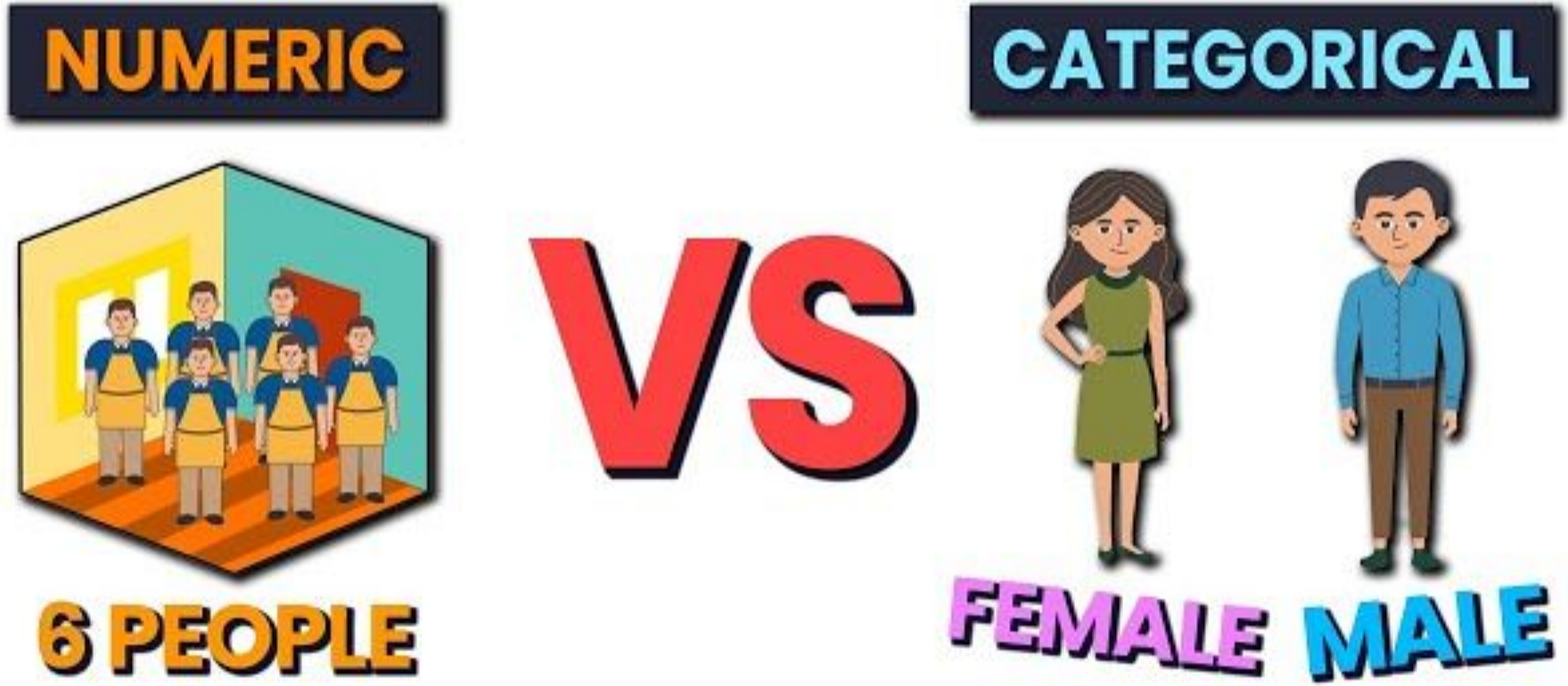
- *Examples:* Industry Sector (FinTech, GreenTech), Subscription Tier (Basic, Pro, Enterprise).
- *Best Visuals:* Frequency Tables, Bar Charts, Pie Charts.

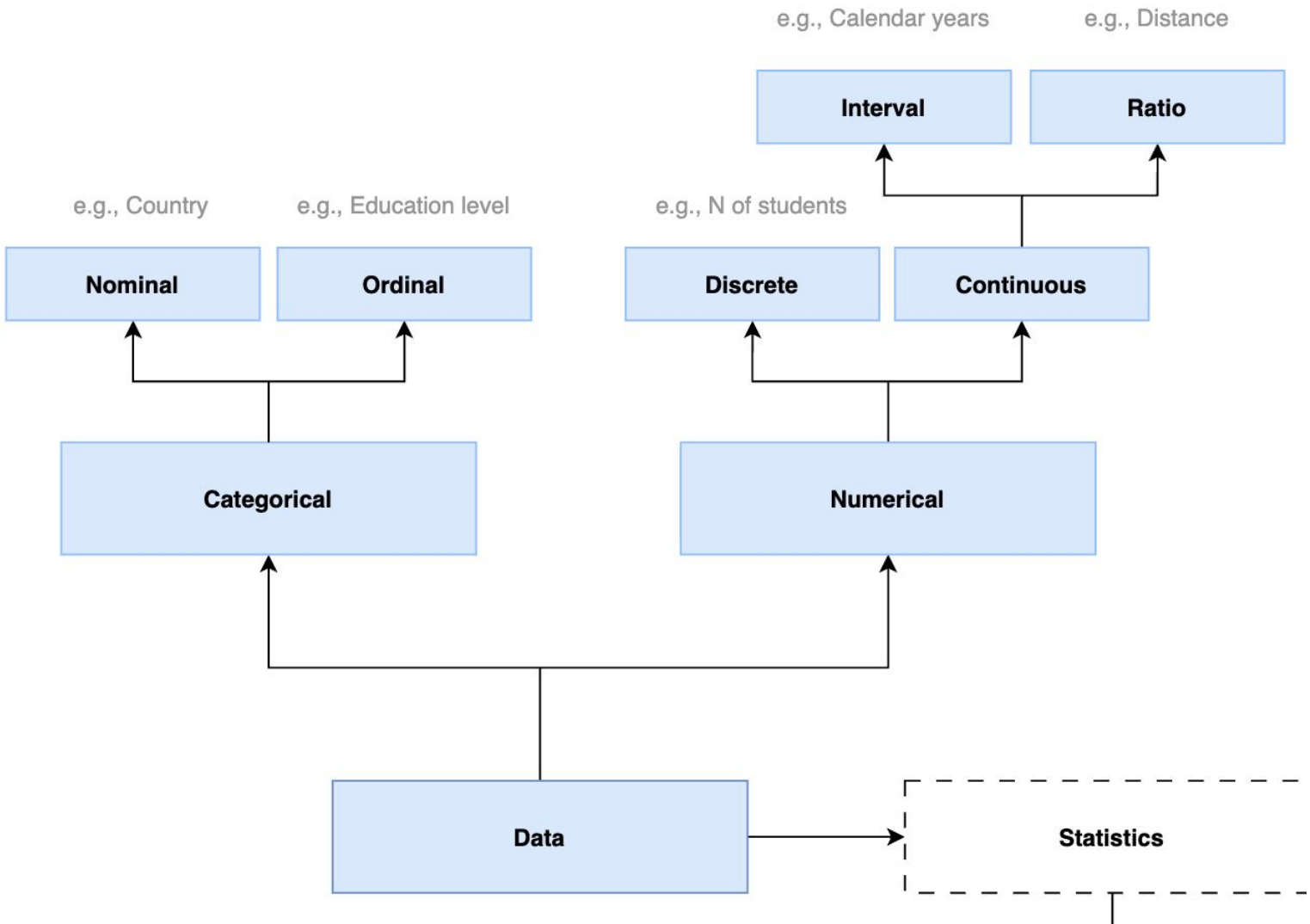
Numerical Data: Quantitative measurements.

- *Examples:* Annual Revenue (Euros), Number of Employees, Customer Age.
- *Best Visuals:* Histograms, Boxplots, Scatter Plots.

The Rule: You cannot use the same chart for every type of data!

Know Your Data Types Before You Code




e.g., Calendar years
e.g., Distance
Interval
Ratio
e.g., Country
e.g., Education level
e.g., N of students
Nominal
Ordinal
Discrete
Continuous
Categorical
Numerical
Data
Statistics

Know Your Data Types Before You Code

Nominal: names only (e.g., Country)

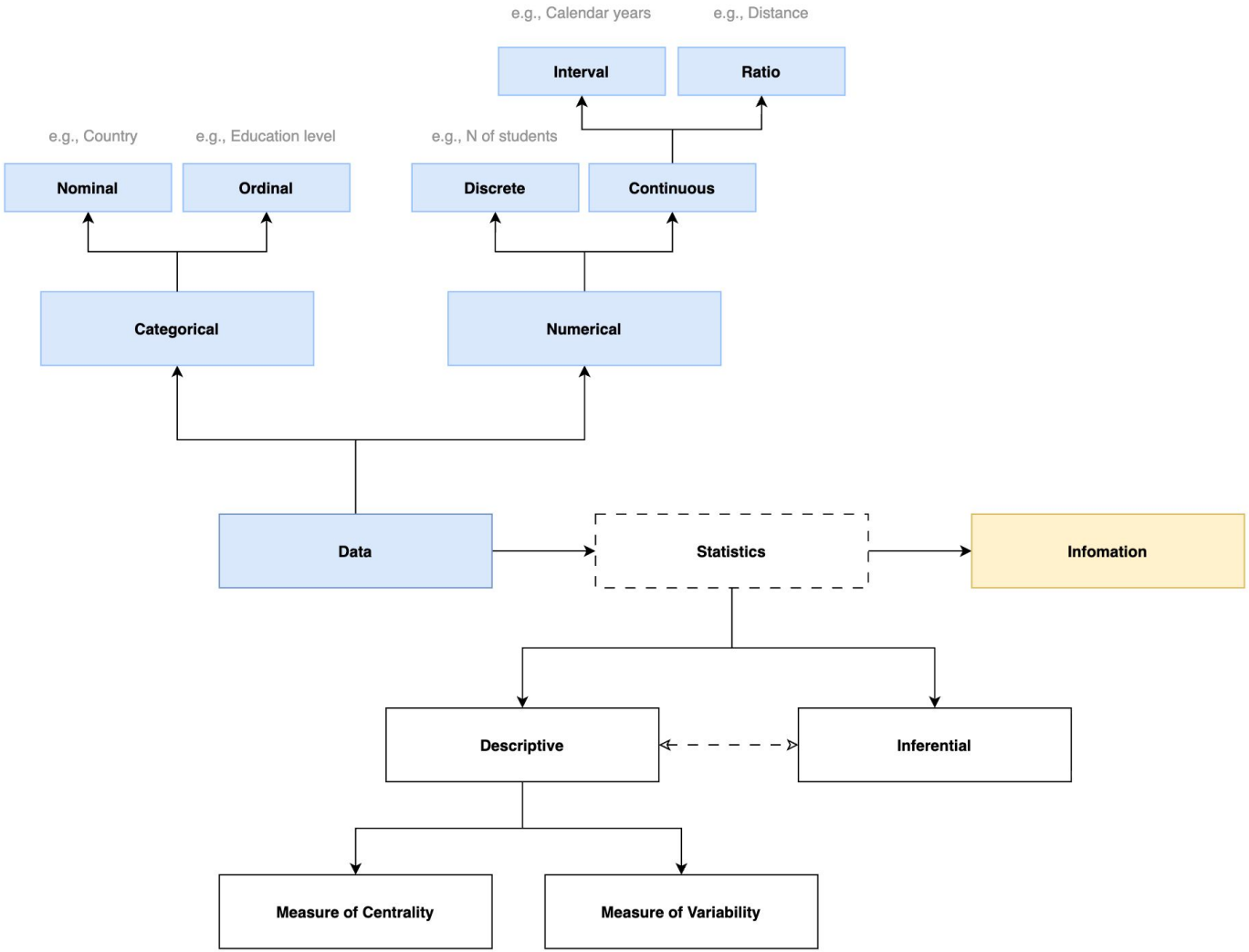
Ordinal: order only (e.g., Educational level)

Discrete: countable numerical values (e.g., Number of students)

Interval: order + meaningful differences + no true zero (e.g., Calendar years)

Ratio: order + meaningful differences + true zero (e.g., Distance)

measurable numerical values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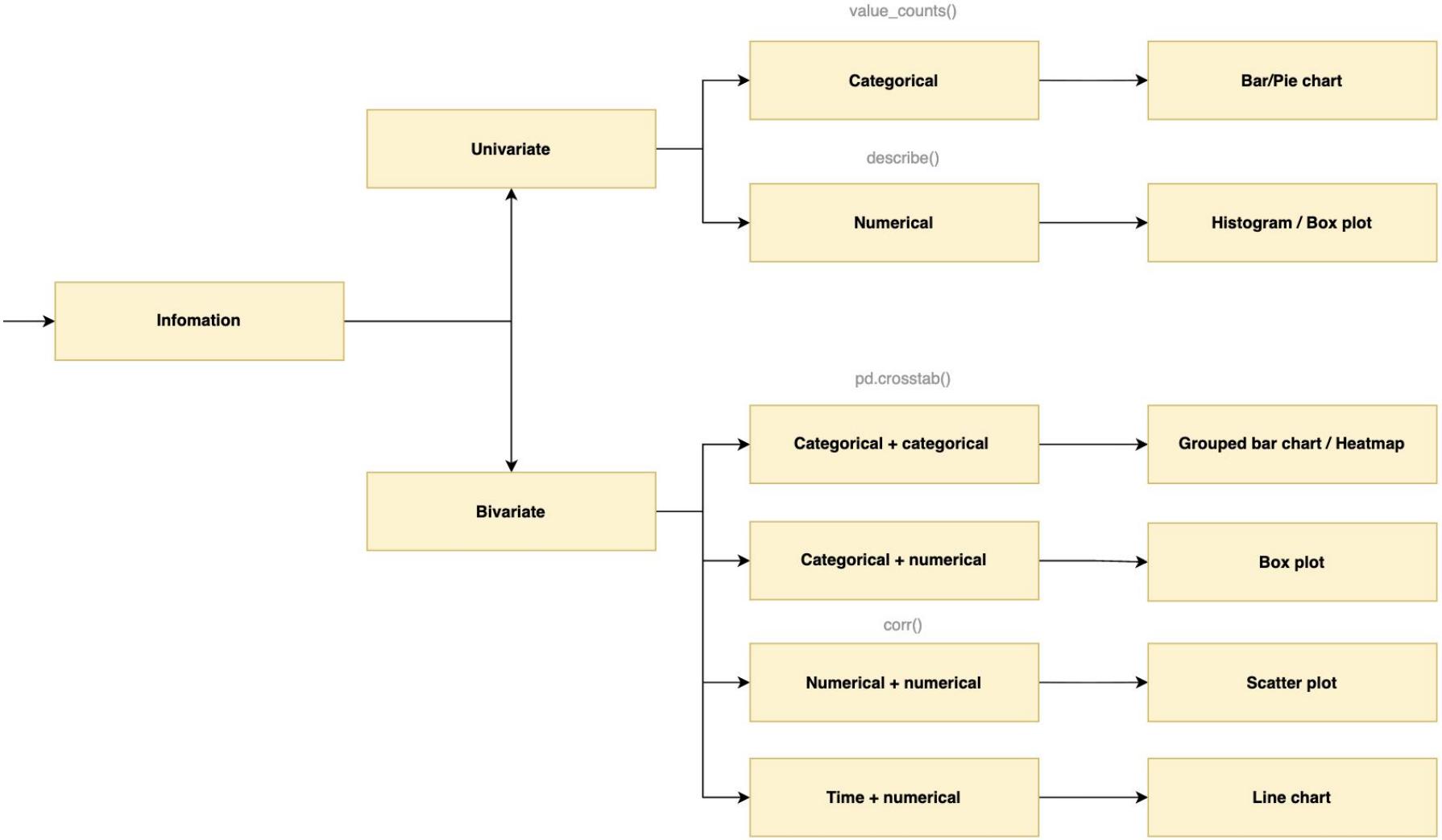
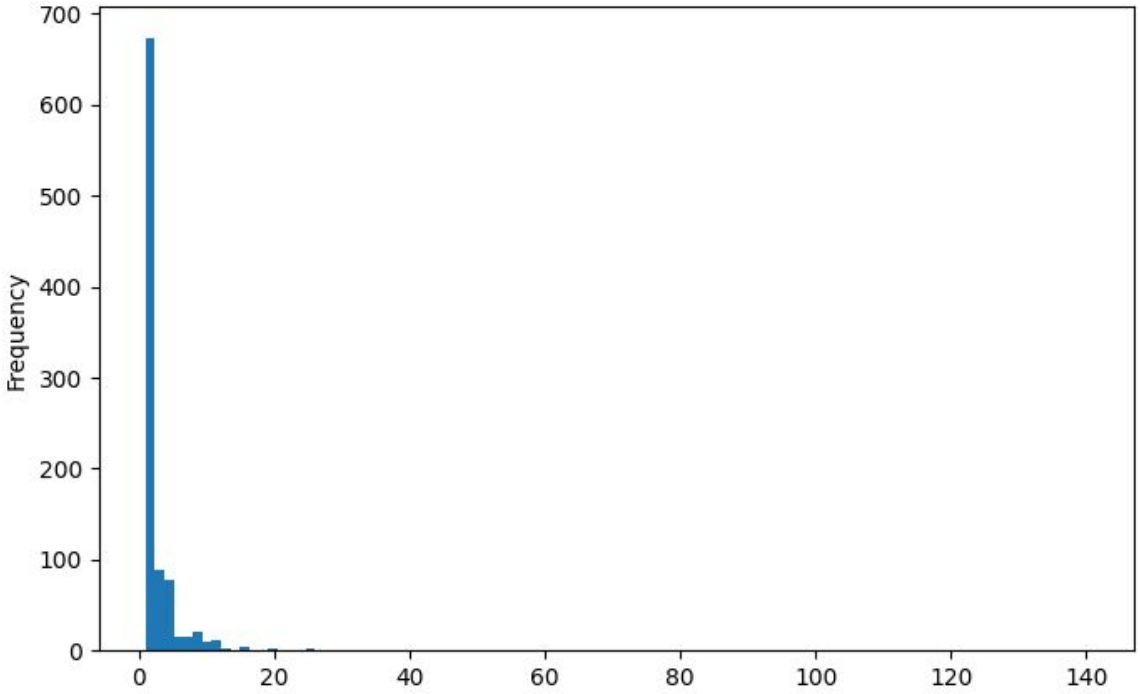

value_counts()
Categorical
Bar/Pie chart
Univariate
describe()
Numerical
Histogram / Box plot
Infomation
pd.crosstab()
Categorical + categorical
Grouped bar chart / Heatmap
Bivariate
Categorical + numerical
Box plot
corr()
Numerical + numerical
Scatter plot
Time + numerical
Line chart

Univariate - Numerical

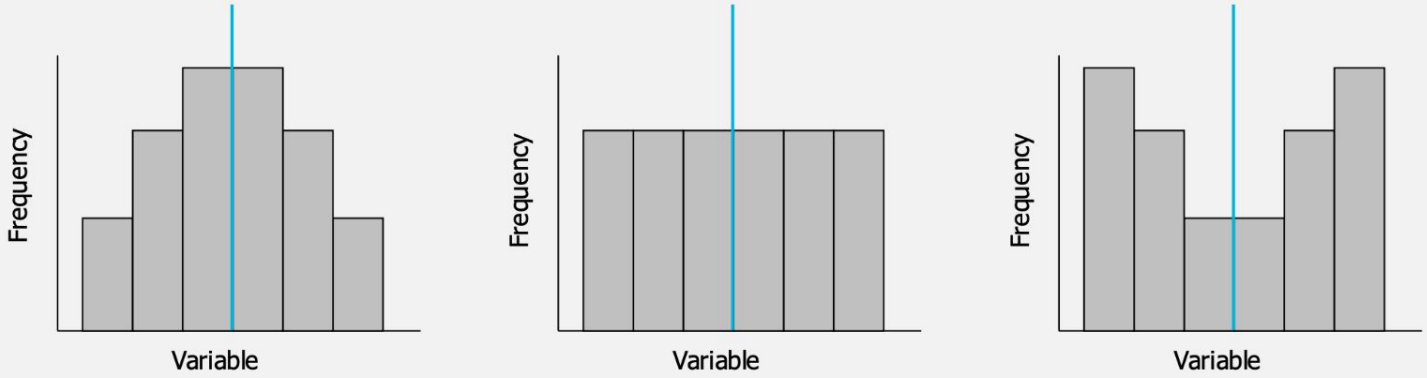
Let's explore the distribution of startup valuations!



SHAPES OF HISTOGRAMS

SYMMETRY

A histogram is said to be ***symmetric*** if, when we draw a **vertical line** down the center of the histogram, the two sides are identical in shape and size:



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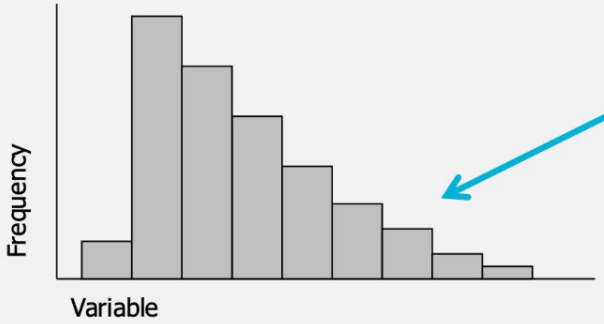
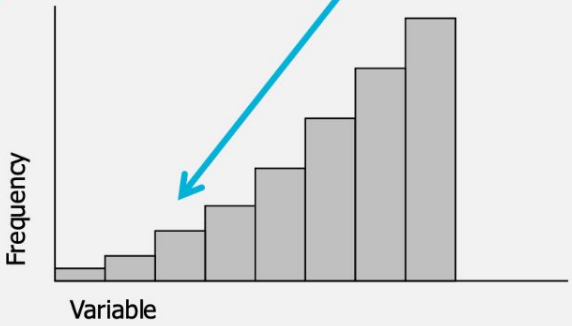
SHAPES OF HISTOGRAMS

SKEWNESS

A skewed histogram is one with a long tail extending to either the **right** or the **left**:



Positively Skewed



Negatively Skewed

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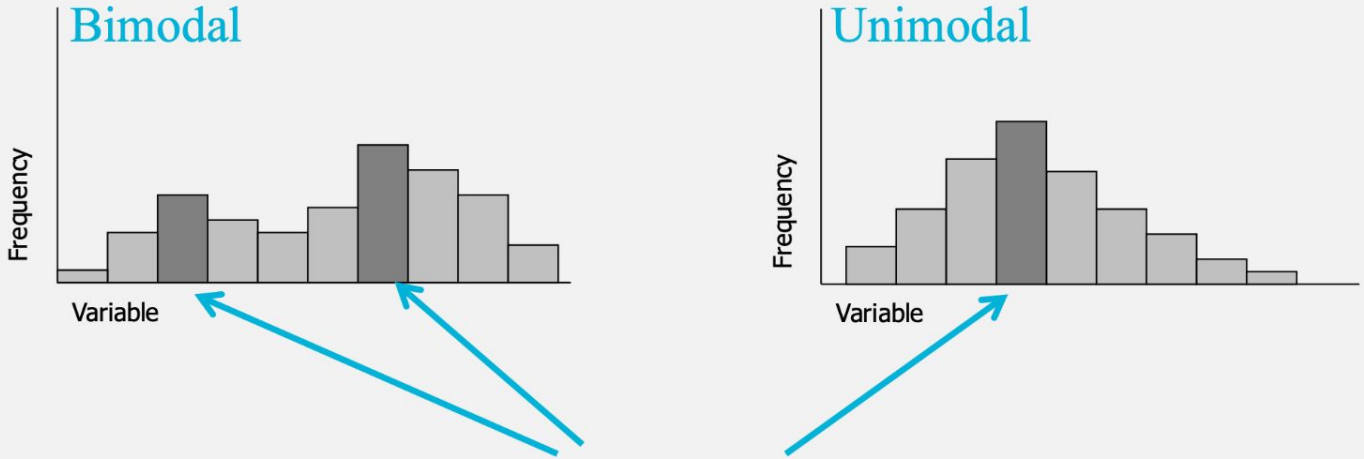
SHAPES OF HISTOGRAMS

MODULARITY

A unimodal histogram is one with a single peak, while a bimodal histogram is one with two peaks:



A **modal class** is the class with the largest number of observations

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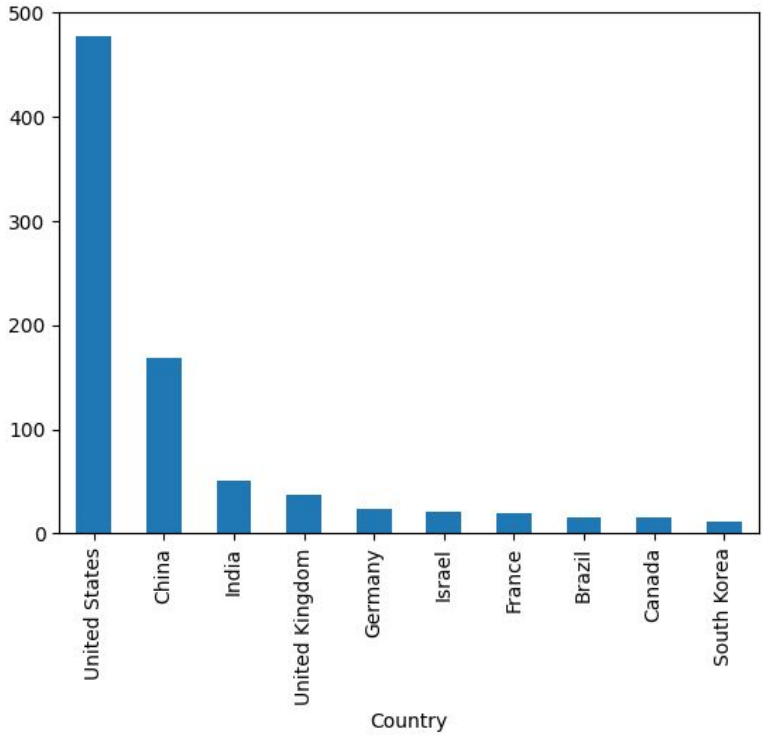
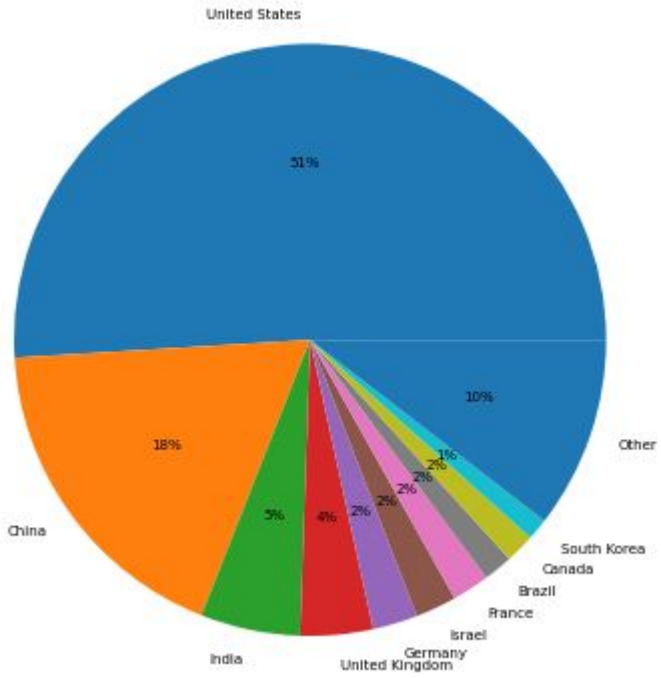
Univariate - Categorical

Let's find out which countries have the most startups!

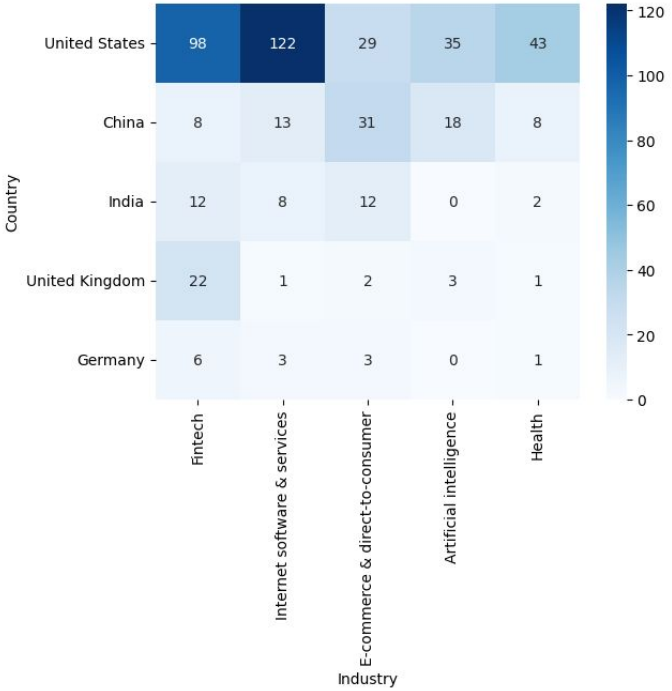




Bivariate: Categorical + Categorical

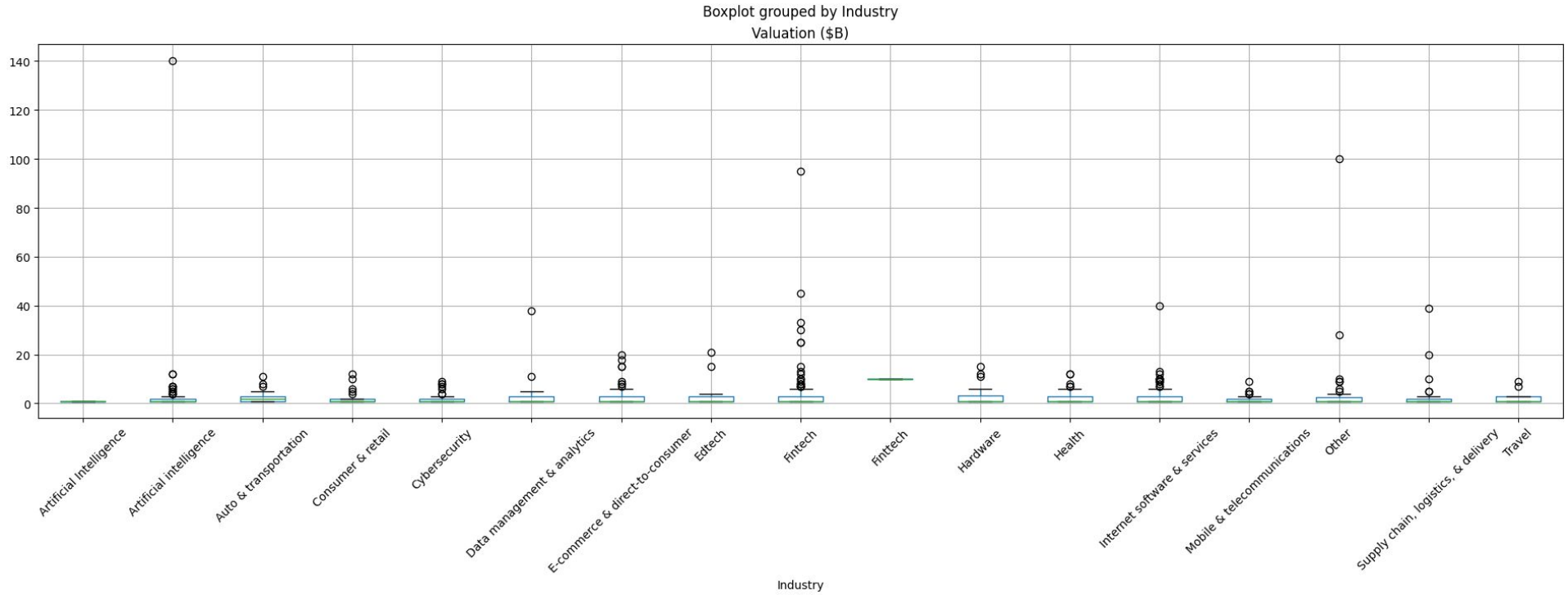
*A **contingency table** (also known as a **cross tabulation** or **crosstab**), also known as a **cross-classification table**, describes the relationships between two or more categorical variables.*

How many Artificial Intelligence startups are located in China?



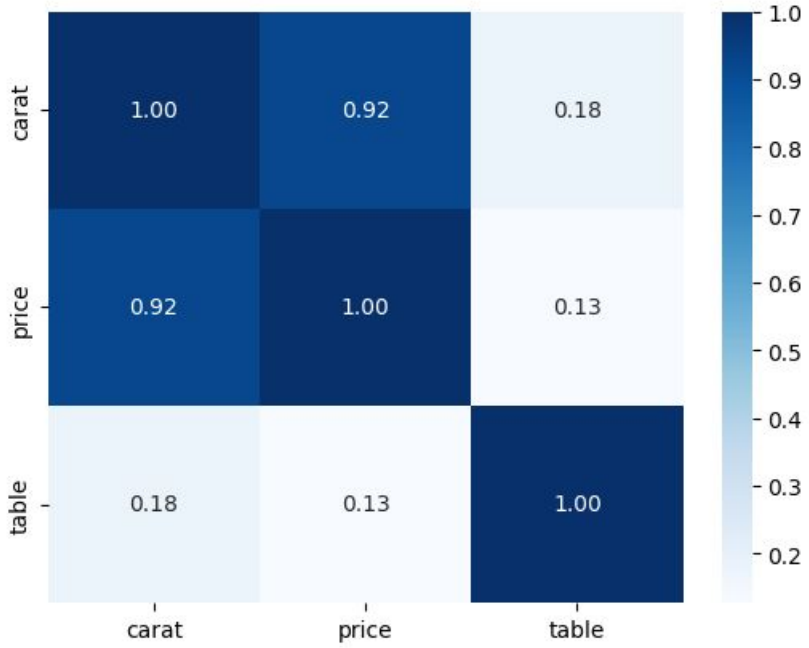
Bivariate: Categorical + numerical

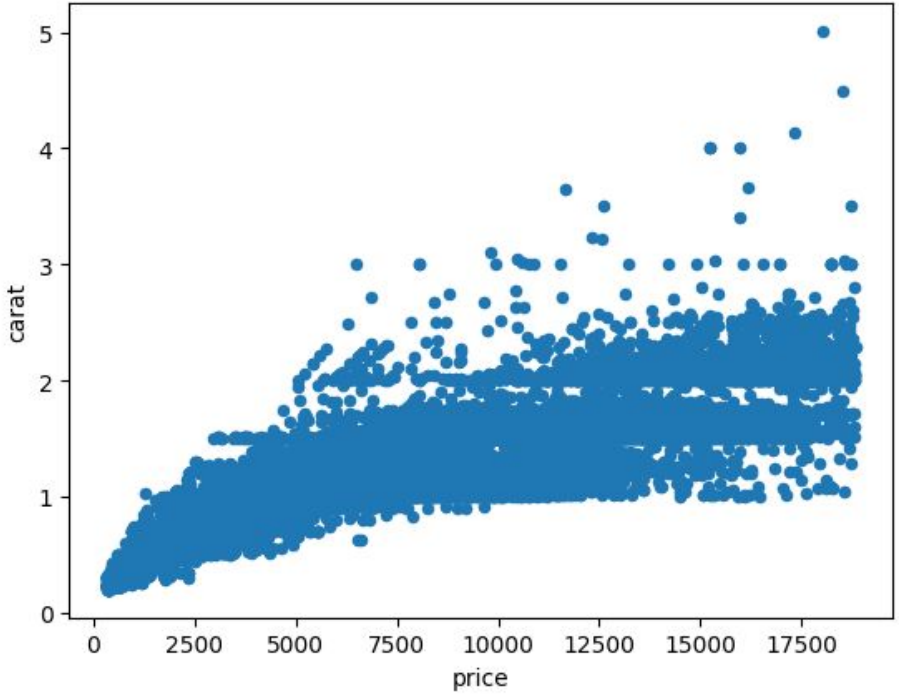
Let's figure out which industry holds the most valuable companies!



Bivariate: Numerical + numerical

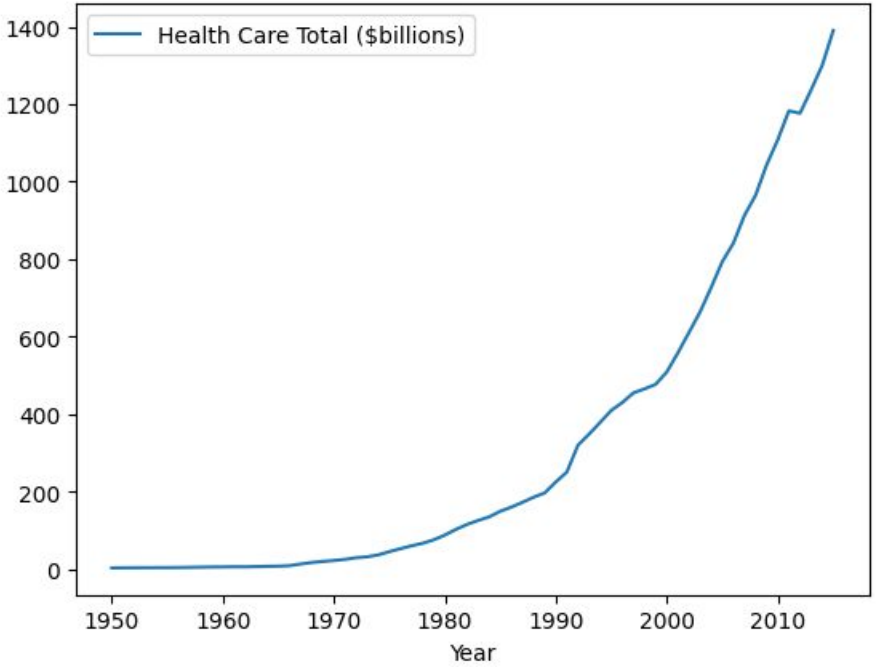
Let's figure out which industry holds the most valuable companies!

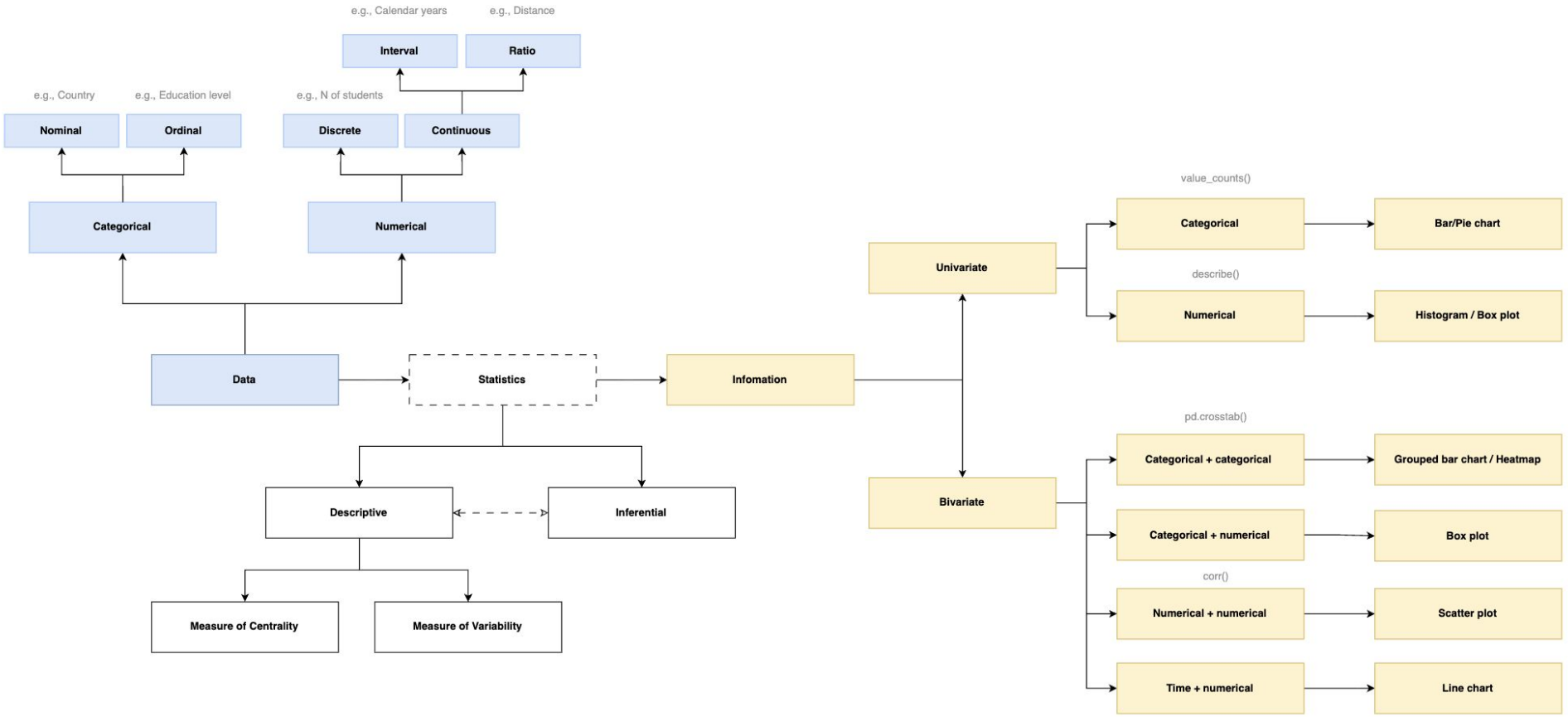




Bivariate: Time + numerical

Look at the period between 1950 and 1970. How would you describe the growth of healthcare spending during that time compared to the years after 1990?

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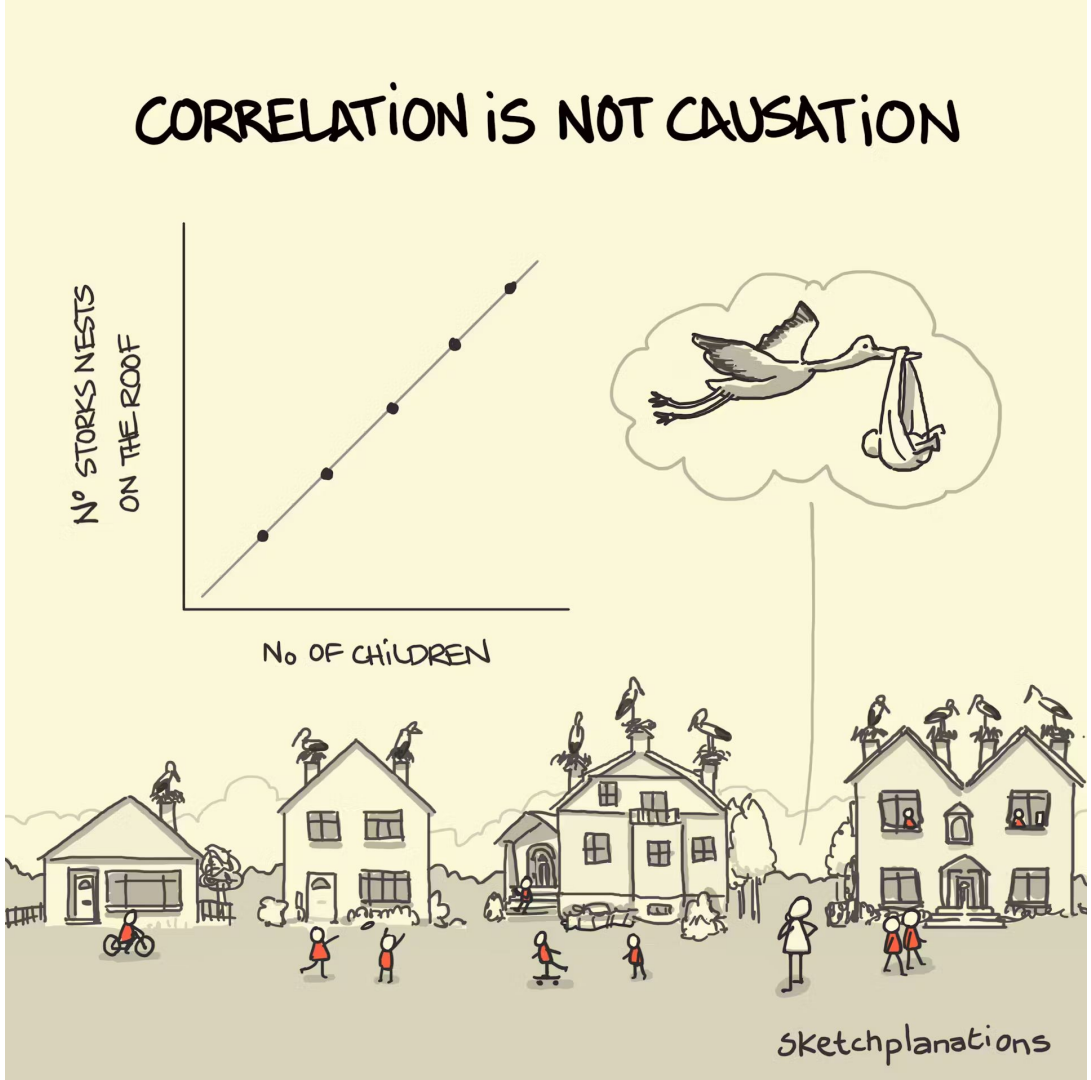
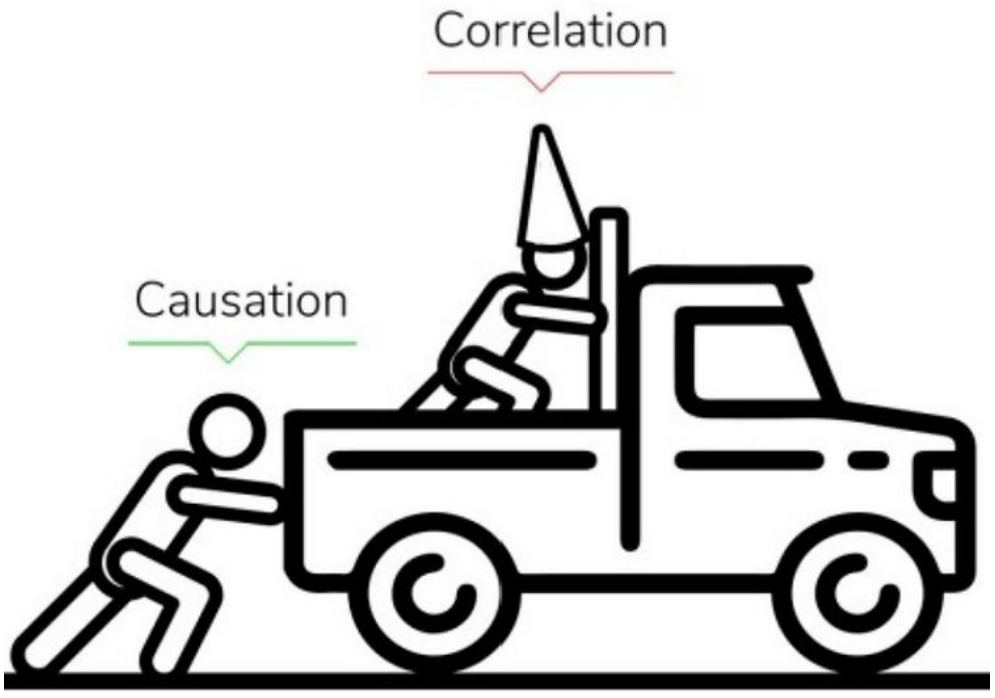

CORRELATION IS NOT CAUSATION
Nº STORKS NESTS ON THE ROOF
No OF CHILDREN
sketchplanations

STORKS

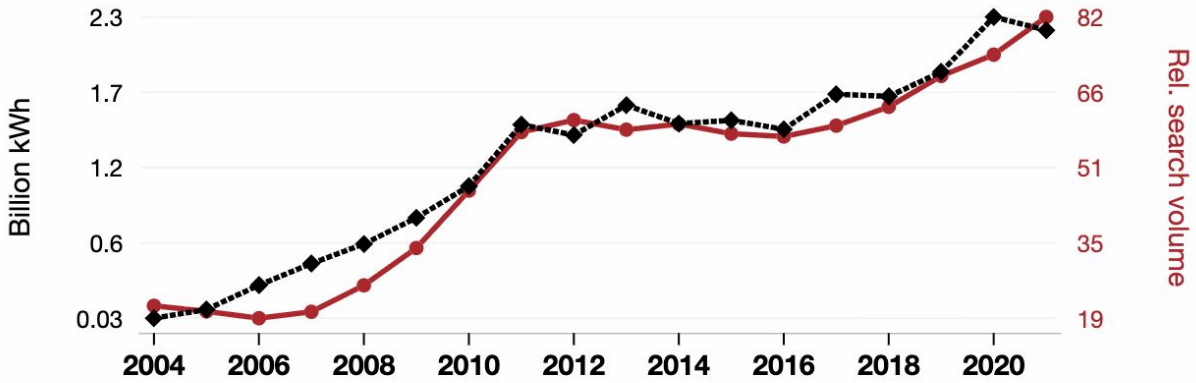
Correlation is not causality



Wind power generated in Taiwan

correlates with

Google searches for 'i am tired'



Total wind power generated in Taiwan in billion kWh · Source: Energy Information Administration

Relative volume of Google searches for 'i am tired' (Worldwide, without quotes) · Source: Google Trends

2004-2021, r=0.980, r^2=0.961, p<0.01 · tylervigen.com/spurious/correlation/1775

View details about correlation #1,775

Show scatterplot

What else correlates?

Wind power generated in Taiwan · all energy

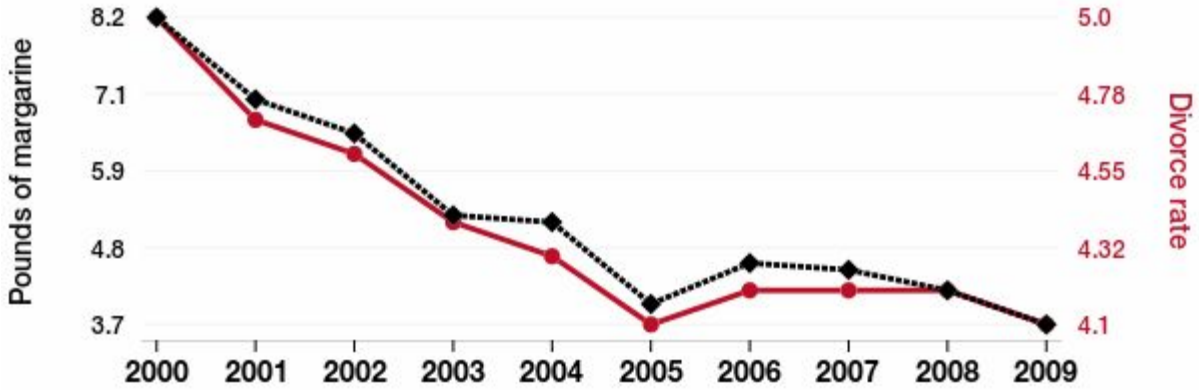
Google searches for 'i am tired' · all google searches

https://www.tylervigen.com/spurious-correlations

Per capita consumption of margarine

correlates with

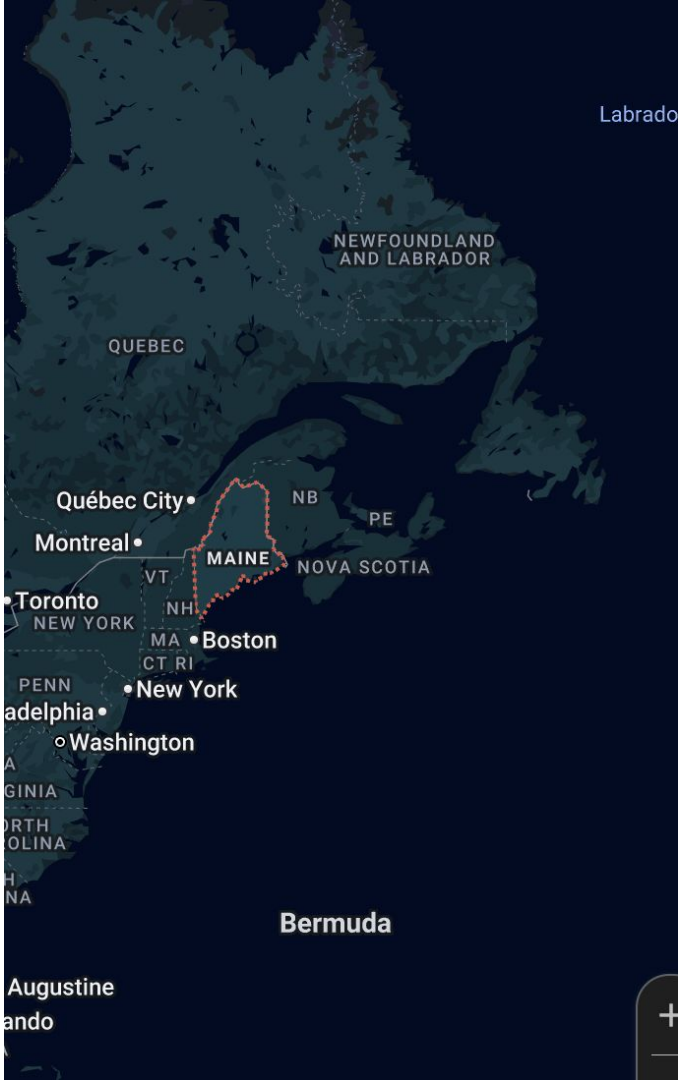
The divorce rate in Maine



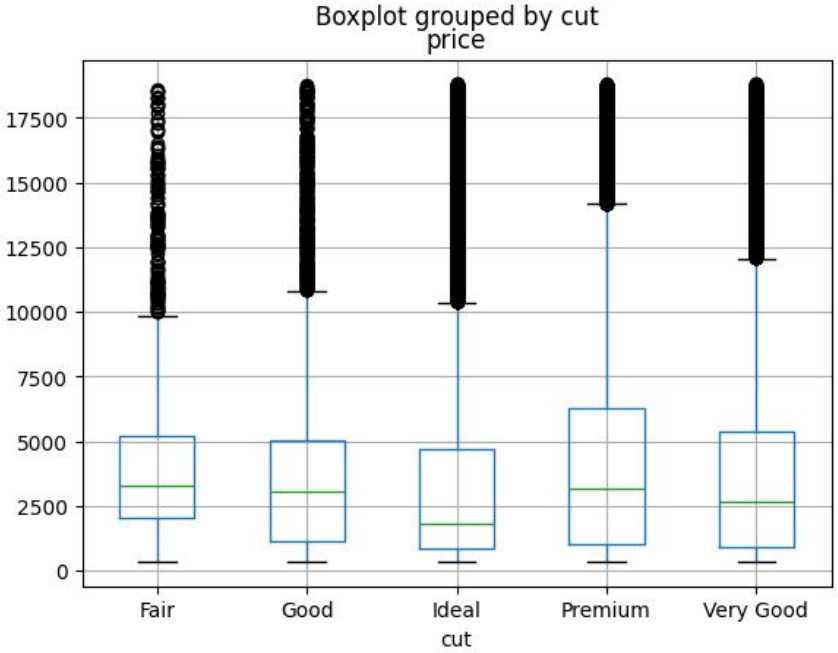
Per capita consumption of margarine in the United States · Source: US Department of Agriculture

The divorce rate in Maine · Source: CDC National Vital Statistics

2000-2009, r=0.993, r^2=0.985, p<0.01 · tylervigen.com/spurious/correlation/5920

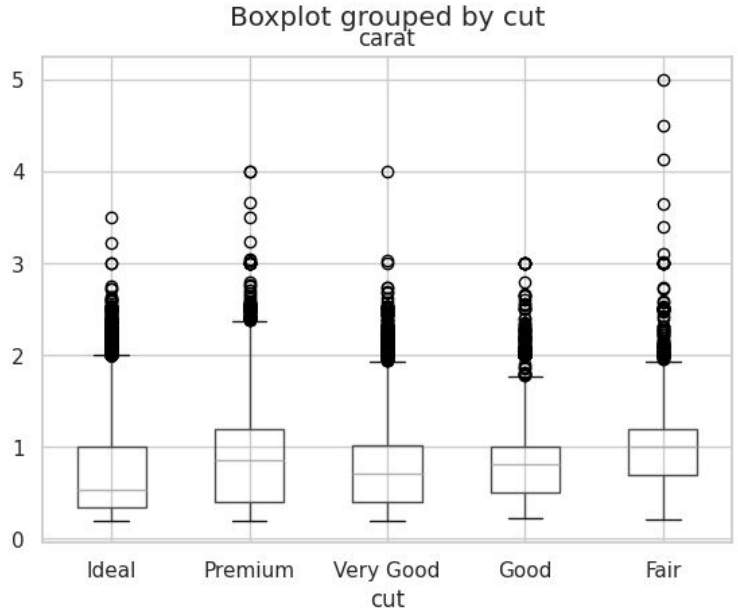


A Data Mystery: Do Worse Diamonds Cost More?



The Trap of Causality (Correlation ≠ Causation)

Remember how our last chart showed that 'Fair' cut diamonds had the highest median price? It didn't make sense.



Scientific Correspondence | Published: 13 May 1999

Myopia and ambient lighting at night

Graham E. Quinn, Chai H. Shin, Maureen G. Maguire & Richard A. Stone

Nature **399**, 113–114 (1999) | Cite this article

4911 Accesses | **138** Citations | **65** Altmetric | Metrics

Abstract

Myopia, or short-sightedness, occurs when the image of distant objects, focused by the cornea and lens, falls in front of the retina. It commonly arises from excessive postnatal eye growth, particularly in the vitreous cavity. Its prevalence is increasing and now reaches 70-

Brief Communication | Published: 09 March 2000

Vision

Myopia and ambient night-time lighting

Karla Zadnik, Lisa A. Jones, Brett C. Irvin, Robert N. Kleinstein, Ruth E. Manny, Julie A. Shin & Donald O. Mutti

Nature **404**, 143–144 (2000) | Cite this article

2243 Accesses | **70** Citations | **34** Altmetric | Metrics

Abstract

Myopia is a common affliction (one in four adult Americans is near-sighted[1]), and juvenile-onset myopia is believed to be due to a combination of genetic and environmental factors[2]. Results from animal experiments indicate that light cycles may affect the development of

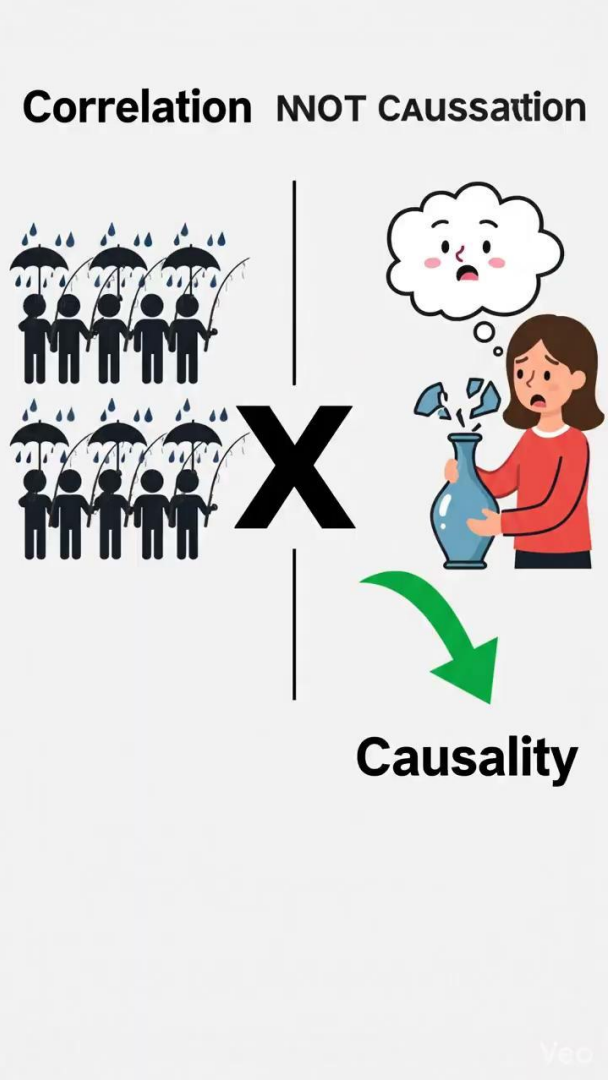
What they found:

The real cause of the children's nearsightedness was simply genetics[6]. Myopic (nearsighted) parents are highly likely to have myopic children[6].

Crucially, **nearsighted parents have poor night vision**. Therefore, myopic parents were significantly more likely to leave a nightlight or hallway light on in the nursery so they wouldn't trip over furniture when checking on their babies in the middle of the night[6]. Parents with perfect vision didn't need the lights, so they left their babies in the dark.

The Conclusion

The nightlight didn't *cause* the nearsightedness. The parents' genetics caused the children's nearsightedness, and the parents' genetics *also* caused the presence of the nightlight[6].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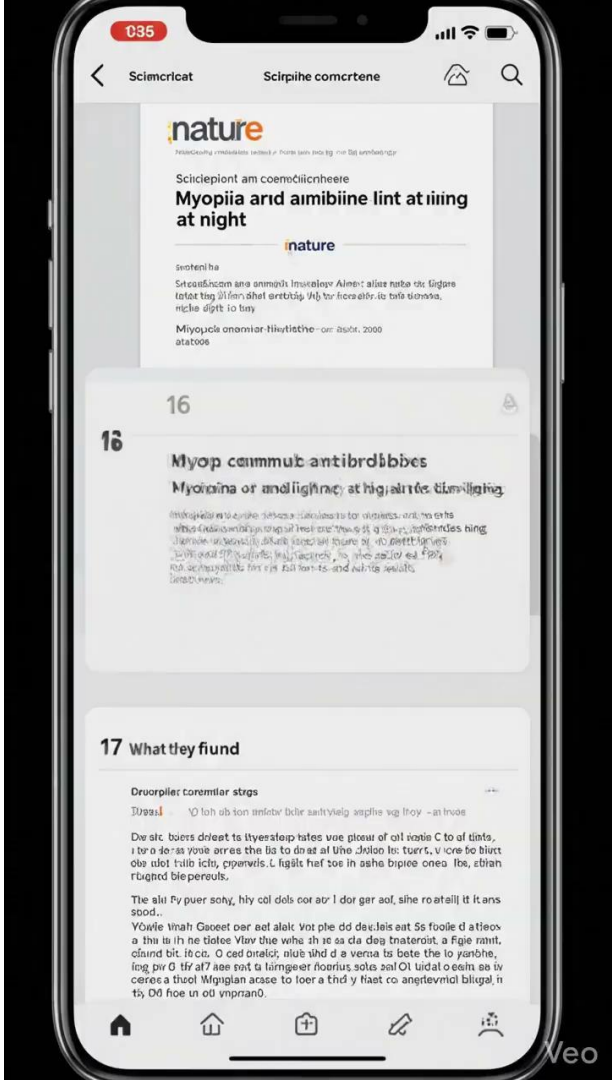
X
Causality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